

Education

University of California, Santa Cruz

September 2013 – June 2017

Bachelor of Science with Honors in Computer Engineering with a focus on Digital Hardware.

Significant Course Work:

- Microprocessor System Design
- Computer Architecture
- Digital Signal Processing
- Logic Design with Verilog
- Signals and Systems
- Analog Circuits
- VLSI Digital System Design
- Logic Design
- Intro to Circuits
- Algorithms and Abstract Data Types
- Data Structures
- Computer Systems and C Programming
- Computer Systems and Assembly Language
- Technical Writing

Experience

Generac Clean Energy

September 2023 – May 2025

Staff Firmware Engineer

Remote

- Developed embedded control systems for managing PLC-based communications with microinverters via a Yitran PLC chip.
- Designed and implemented automatic set/verify routines for grid profiles, max power limits (including PTO states), and microinverter calibration/control parameters.
- Created robust CLI and CAN-based interfaces to enable full microinverter control, diagnostics, and calibration from external systems.
- Built a protocol translation layer to map SunSpec data models to a proprietary microinverter protocol, supporting DER compliance.
- Participated in and supported IEEE 2030.5 (CSIP) compliance testing, ensuring protocol alignment and interoperability.
- Engineered a grid phase-lock mechanism based on voltage thresholds, replacing zero-cross detection and significantly improving PLC stability on noisy grids.
- Designed a next-generation remote control P2P protocol that re-implemented GUI objects on the client side (Java) rather than transmitting raw screen captures—improving responsiveness from 10s/frame to ~0.5s/frame.

SunPower Corp

November 2018 – September 2023

Staff Firmware Engineer

Remote / Richmond, CA

- Lead engineering efforts to work around supply chain constraints in order to ensure product supply continuity – determined alternate components and implemented manufacturing and production support.
- Maintained product supply throughout the supply chain crisis by developing creative solutions for alternate hardware designs and components.
- Lead team of 5 developers for legacy product sustaining work. This included designing creative solutions for reducing flash degradation and developing support for IEEE1547-2018 regulatory requirements.
- Architecture design and BSP bring-up for the next generation SunVault (energy storage) product.
- Developed firmware that enabled and improved communications between the PV supervisor (PVS6) and the devices it monitors/controls. Heavy focus on Enphase IQ7 micro inverters communications and support.
- Worked with utilities to support regulatory requests such as HECO, Rule21/IEEE2030.5, and IEEE1547 based grid profiles.
- Brought up firmware for municipal interconnect disconnect controller (MIDc) which manages the grid disconnect contactor switch for SunVault systems.
- Network stack support for BG95/96 modules.

Q Analysts Test Engineer onsite at Facebook HQ
Test Technician (Lab Data Collection, Scripting)

March 2018 – Present
Menlo Park, CA

- Developing automation software for the QA robotics lab which includes a Yaskawa 6-axis robotic arm, Philips Hue lights, and virtual reality (VR) headsets.
- Overseeing experiments run in the lab (including setup, configuration, executing, and basic data analysis)
- Extracting, processing, and analyzing CSV and JSON data files produced from files on the headset.

Computer Science Database Systems II Staff
Tutor and Grader

April 2017 – June 2017
UC Santa Cruz

- Graded assignments for Professor Sheldon Finkelstein's Database Systems CS capstone course.
- Tested student's implementation of database systems using C++ test benches.
- Assisted students with questions regarding C++ and course concepts.

RT-2M Replacement Senior Design Project

January 2017 – June 2017

Lead DSP Engineer

Plantronics

- Interdisciplinary senior design project that maintained a professional relationship with Plantronics through biweekly progress evaluations.
- Designed LabVIEW software for the NI cRIO to replace Plantronics' outdated audio test equipment.
- Utilized event driven, object oriented, and multithreaded LabVIEW code for a robust software architecture.
- Engineered Digital Signal Processing algorithms for crest factor optimization, calibration, and audio signal generation and analysis.
- Successfully provided Plantronics with a working prototype of an RT-2M replacement tester.
- Worked in a team of six.

Leeps Lab Research Intern

December 2015 – December 2016

Lead Software Engineer

UC Santa Cruz

leeps.ucsc.edu

- Worked under Professor Kristian Vargas Lopez on a behavioral economics project.
- Developed a facial recognition program to determine and log the emotional state of a subject.
- Programmed using Affdex, Boost, and OpenCV C++ SDKs in a Linux environment.
- Utilized Shimmer Sensors to log heart rate (PPG) and skin conductance (GSR) data.
- Configured pilot experiments involving collecting data from student volunteer subjects.

Projects

Sit/Stand Desk Personal Project

August 2017 – November 2017

- Designed and built a desk that can move up and down using an Arduino based embedded system that controls two linear actuators and a user interface (LED buttons and an LCD screen).
- Programmed a manual mode for the user to control the desk, an automatic mode where the user can specify a time interval for the desk to alternate between two heights, and the ability to save presets.
- Filtered out unwanted RF produced by the motors in the linear actuators.

RISC V CPU Synthesis Project

January 2016 – March 2016

Class: Logic Design with Verilog

UC Santa Cruz

- Synthesized a RISC V instruction set architecture CPU using Verilog.
- Designed using a three-stage fluid pipeline (fetch, decode, and execute) for better throughput.
- Implemented C++ testbenches to mimic an operating system for the CPU.
- Verified functionality using GTKWave and Yosys.

WalkVR - Hackathon Project

October 2015

Lead Map Developer

CalHacks2.0

- Developed a 3D environment using Unreal Engine for testing the features of our VR hardware.
- Used an Oculus Rift DK2, Leap Motion, and Myo Armband to create a full VR experience.
- Leap Motion tracked hand gestures while the Myo Armband was worn around the ankle to track walking.
- Worked in a team of five.

ObjectRekt – Hackathon Project
Embedded Software Developer

June 2015
Flir Hackathon

- Utilized the Flir Lepton longwave infrared thermal imager, Raspberry Pi, and OpenCV to create an automated camera that observes the scene and tracks a presenter's location, panning to the proper locations.
- Developed C++ program that determines the presenter's subject (whiteboard, projection, etc.) using an object recognition and then, using the Lepton Python SDK, the system recognizes the presenter's location and pans the camera accordingly to accommodate both the presenter and their subject.
- Worked in a team of five.

SlugTrails – Hackathon Project
Software Developer

January 2015
HackUCSC

- Developed an Android and IOS app with the purpose of crowdsourcing wildlife sightings.
- Implemented the Google Maps SDK to allow the user to tag their location with a time, animal, and a description of the sighting. This data was added to a database.
- Tested app using data generated from a Python script
- Taught myself C# for developing apps on Apple devices.
- Worked in a team of five.

Emocar - Sponsor Prize at CalHacks Hackathon
Lead Embedded Software Engineer

October 2014
CalHacks

- Designed a brain-computer interface that allows a user to control an Arduino rover with their mind.
- Utilized a machine learning algorithm to detect patterns in noisy data. Used this to determine if the raw EEG data matched a command for the rover.
- Won a sponsor prize for most connected project.
- Worked in a team of four.

Hartbeat - Alpha Game Jam Project
Lead Hardware Engineer & Lead Map Developer

September 2014
Alpha Game Jam

- Engineered an embedded system that captured the user's heartrate using an optical heartrate sensor and a pulse width modulation algorithm.
- Utilized the heartrate data to affect bullet spread radius to a first-person shooter testing environment we developed in UDK and UnrealScript.
- Developed map environment and heads-up display graphics.
- Worked in a team of six.

FindAR - First Place at HeroHacks Hackathon
Software Engineer

August 2014
Hero Hacks

- Engineered an augmented reality headset by mounting a camera on an Oculus Rift DK1.
- Designed computer vision filters using C++ and OpenCV including: filter by color, facial recognition, object detection, edge detection, and other various color filters.
- Worked in a team of five.

Skills

10yrs: Embedded Systems: (STM32, IMX6/8, PSoC, Atmel, RTOS/FPGA, NI cRIO, Xilinx FPGA, Arduino)

10yrs: Git for large scale version control and project development.

4yrs: CI/CD (Yocto, GitHub Actions) for building projects.

4yrs: DER Regulatory Requirements (IEEE2030.5 CSIP, IEEE1547, CA-Rule21, HECO)

4yrs: Inverters (Chilicon Power / PWRMicro, Enphase IQ7, SolarBridge, Delta, SMA)

3yrs: Broadband and Cellular Embedded Support

3yrs: Team Management (Direct hires and contractors)

3yrs: Digital and Analog Circuit Design and Analysis

2yrs: Hardware Synthesis and VHDL (Verilog and System Verilog)

2yrs: Computer Vision

1yr: Digital Signal Processing

Solar Tech:

Enphase IQ7
SolarBridge Mis
Chilicon Power
BMS
MIDC (Storage
Disconnect
Controller)
Metering
Commissioning
IEEE2030.5
IEEE1547
HECO
CA-Rule21

Programming:

U-Boot
C/C++
Python
Bash
Batch
Yocto/Bitbake
Javascript
Java
JSON
XML
ProtoBuf
FlatBuf
Swagger
LabVIEW

Software:

VMWare/VirtualB
ox
VIM
VS Code
Git
PuTTY
MobaXTerm
WireShark
PSoC Creator
Microchip/Atmel
Studio
Xilinx Suite
Unreal Engine
UDK

Device Comms:

UART
I2C
SPI
USB
PLC
CAN
Modbus
Sunspec
AT (for BG95/96)
CGI
RPC
MQTT
AWS IOT
COBS
HSSP (for PSoC4)

Hardware & OS:

STM32
NXP LPC-17xx
IMX6, IMX8
PSoC4 PSoC5
Atmel SAM
NI Crio
Zephyr
FreeRTOS
Windows
Linux (Ubuntu)